Detection of Ranch Stock Ponds and Round Water Tanks

Goal: automatically find every livestock water source on grazing ranches from satellite and aerial imagery, both natural or earthen stock ponds and man-made round troughs and storage tanks, as the first step toward long-term monitoring of whether they still hold water. The final models use a self-supervised ViT-L backbone (DINOv3-sat) with additional domain-adaptive self-supervised training on unlabeled county imagery (Section 7).

7,707
pond footprints county-wide (self-supervised ensemble, union of NAIP 2018/2020/2022)

6,735 → 3,072
ponds holding water, autumn 2018 → summer 2022 (severe drought)

F1 0.845 • IoU 0.905
pond detection on the test area (original model 0.833 • 0.829); recall of water-holding NWI ponds 81%

3,026
round tanks / storage tanks county-wide (original model 1,849); test-area recall 78%, precision 60–87%

Data: USDA NAIP, USFWS NWI, Census TIGER, Microsoft Building Footprints, TWDB well databases, Google satellite imagery (tank detection only), Esri World Imagery (visual verification).
Analysis: 19–22 October 2026 • Code and data: server `/home/ychen289/whe/pond`

1. Summary

- **Two kinds of water source, two model branches.** Ponds are segmented in public NAIP aerial imagery (0.6 m, with near-infrared). Round tanks are mostly 2–12 m across, only a few NAIP pixels, so they are detected in Google satellite imagery (~0.25 m) and their type is verified on ~0.13 m chips.
- **Pond detection works.** On a held-out test area (~650 km²), a manual check of 60 random detections gave 88% (strict) to 93% (lenient) precision. Of 60 NWI-registered ponds, 42 held water that year and the model found 34, a recall of 81%. False positives are mostly asphalt roads and dark roofs; misses are mostly small water under tree shade and algae-covered ponds.
- **Year-by-year water tracking is possible.** The same model was applied to 2018, 2020 and 2022 imagery (after radiometric matching of 2018/2020). With the original model, ponds holding water fell from 6,688 to 3,152 county-wide. In 2022 only 17% of ponds under 500 m² still held water, against 82% of ponds over 1 ha.
- **The first round-tank system was usable but limited.** The two-stage system reached 72% recall and 67–85% precision on the test area (1,849 tanks county-wide). The main difficulty is that, at high resolution, round hay bales and hay-ring feeders look almost identical to troughs: only 29% of the detector's high-confidence hits were water facilities. A second-stage classifier fixed most of this.
- **Self-supervised upgrade (Section 7).** All models were replaced by a self-supervised ViT-L (DINOv3-sat, further self-supervised on unlabeled county imagery, where it learned to use near-infrared).
  - **Ponds:** the ensemble raises pixel IoU from 0.829 to 0.905 and F1 from 0.833 to 0.845.
  - **Tanks:** a ViT dense detector plus ViT classifier finds about 30% more real facilities at equal recall, or raises recall to 78–89%.
  - **Few labels:** self-supervision helps most here. Tank classification F1 rises from 0.61 to 0.76 with ~90 labels. For ponds, the domain-SSL model trained on 10% of the imagery (F1 0.831) matches the U-Net trained on all of it (0.833).
- **Main limitations.** Evaluation samples are small (about ±12 percentage points). Google imagery dates are unknown and its terms restrict use, so it is suitable only for locating tanks, not for judging whether they held water on a given date.

2. Study area and data sources

The study area is Hamilton County in central Texas (~2,170 km²). It is mostly grazing land, with many earthen stock ponds (locally called stock tanks), round troughs next to windmills and wells, and metal storage tanks. No ready-made labels existed: all training labels were generated automatically from public data or labeled manually by the author.

Data	Source / access	Resolution • date	Use
NAIP orthoimagery	USDA; downloaded via the Microsoft Planetary Computer STAC API (77 quarter-quads × 3 years, ~100 GB)	0.6 m, RGB+NIR; Oct–Nov 2018, Aug–Dec 2020, Jul 2022	Main pond data; year-by-year water status
National Wetlands Inventory (NWI)	USFWS wetlands map service (ArcGIS REST)	Vector; 5,497 "Freshwater Pond" polygons	Pond location prior (weak labels); recall check
TIGER roads	U.S. Census Bureau, 2023	Vector	Hard negatives (asphalt looks like water)
Microsoft Building Footprints	Planetary Computer (ms-buildings)	Vector; 12,222 buildings	Hard negatives; pool flag
TWDB well databases (SDR + GWDB)	Texas Water Development Board, public download	Points; 292 wells with use "Stock"	Clues for finding round tanks (coordinates and use only)
Google satellite tiles	Google Maps XYZ tiles (permitted by the requester); 520 k z19 tiles county-wide, z20 on demand	z19 ≈ 0.25 m, z20 ≈ 0.13 m; acquisition dates not published	Round-tank detection and type classification
Esri World Imagery	ArcGIS online basemap (WorldView-2, Dec 2021)	~0.5 m	Second view for manual verification

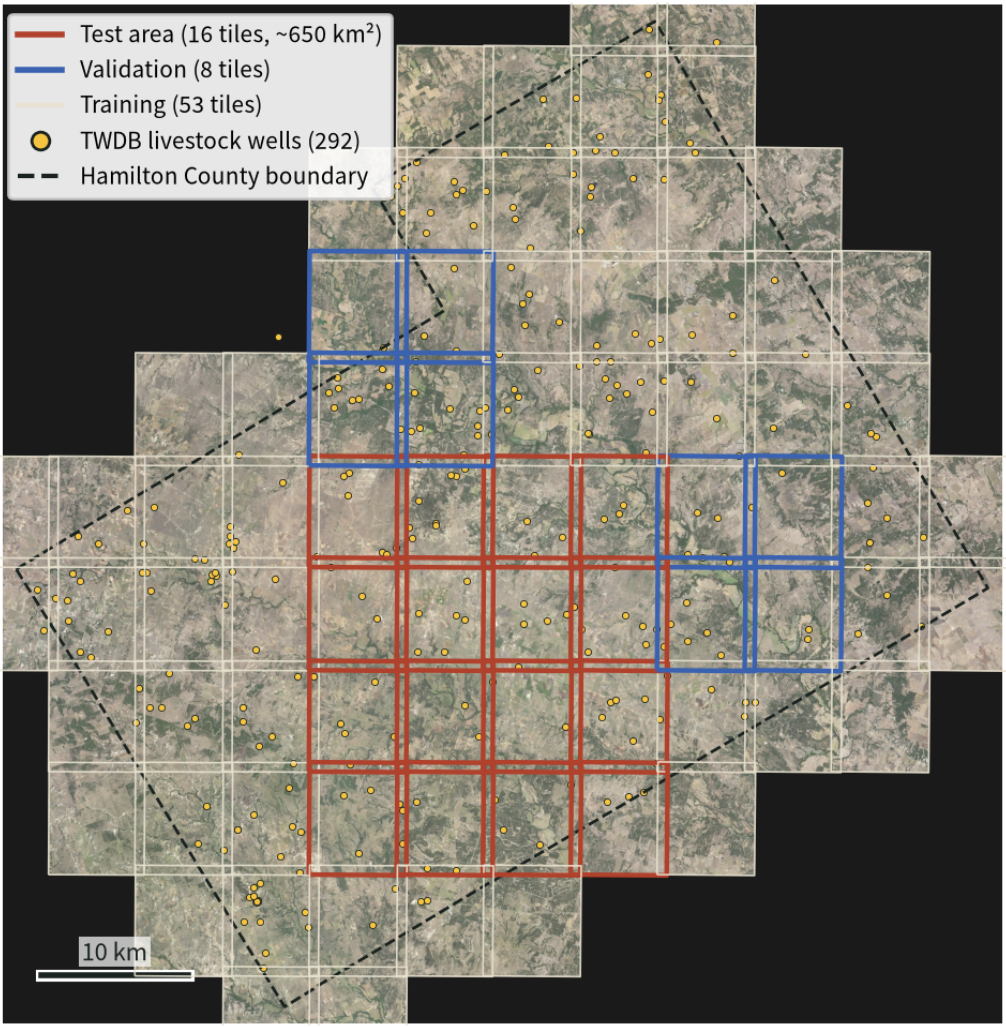


Figure 1. Study area and data split. Background: NAIP 2022 county overview; squares: NAIP quarter-quads. Training, validation and test areas are separated as contiguous spatial blocks. The test area (red) is a block of 16 tiles in the south-central county that no model was trained on. Yellow dots: TWDB livestock wells.

3. Methods

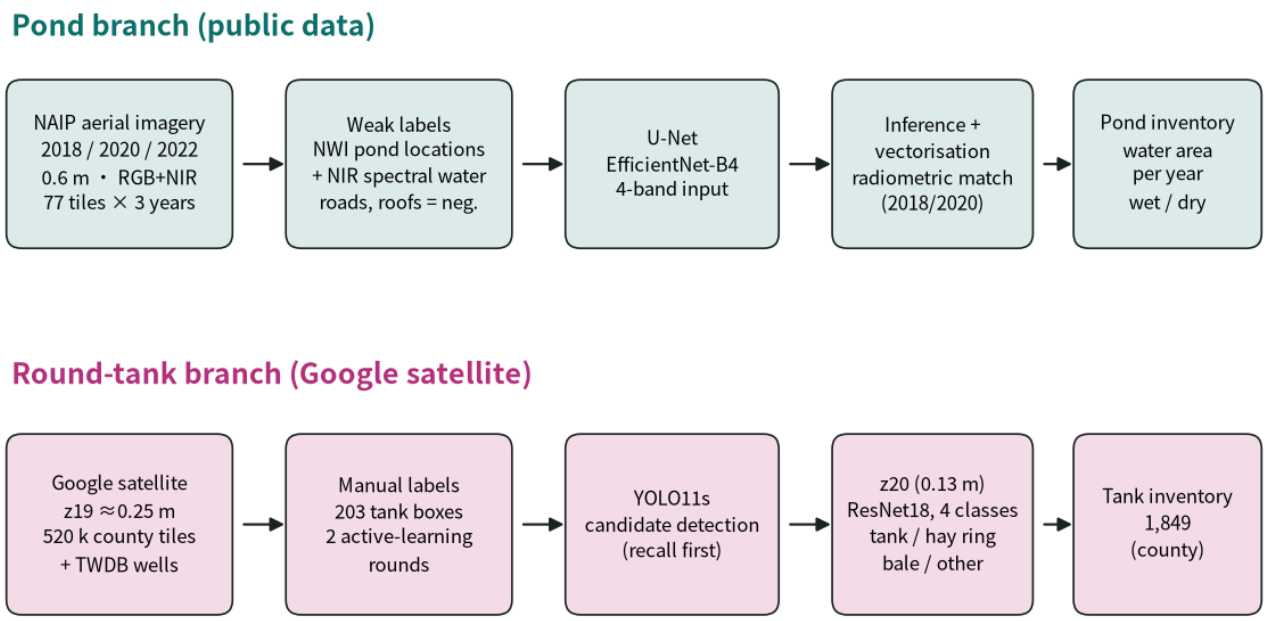


Figure 2. Workflow. Top: pond branch, entirely public data. Bottom: round-tank branch; candidates are detected on z19 imagery and their type is checked on sharper z20 imagery.

3.1 Ponds: weak labels and semantic segmentation

Weak labels. NWI pond polygons are often offset by 20–50 m and many ponds have dried up, so they cannot be used as labels directly. Instead, spectral water is extracted with the near-infrared rule NIR < 0.8×R and NIR < 0.8×G, and each connected component is classified. Components within 60 m of an NWI pond, or compact ones with NDWI > 0.25, are positives. Components on roads or buildings, elongated ones (ditches), and dark low-NDWI tilled soil are negatives. Everything uncertain is set to "ignore" (excluded from the loss). Algae-covered ponds fail the spectral rule, so non-water pixels inside NWI ponds are also ignored, to avoid teaching the model to miss them.

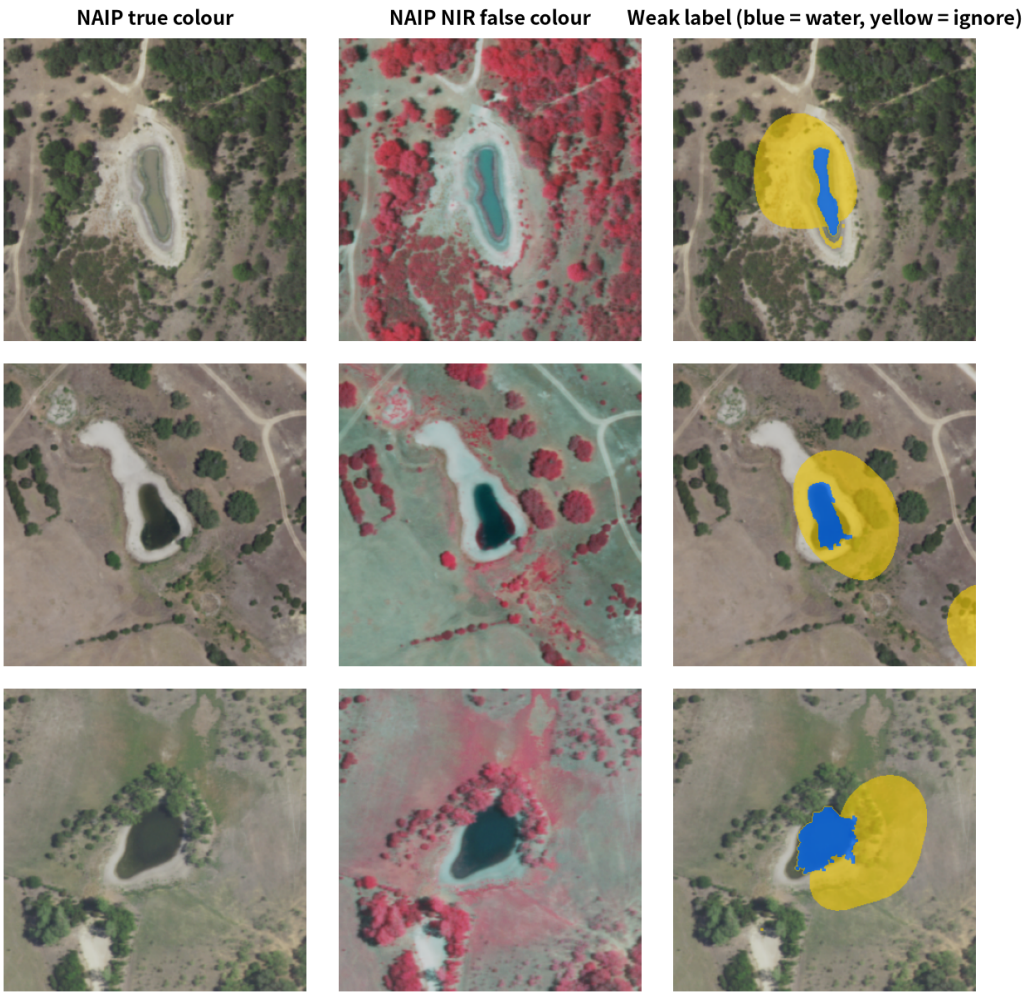


Figure 3. Weak-label examples. Left: NAIP true colour; middle: NIR false colour (water appears dark blue/cyan, vegetation red); right: automatically generated label, blue = water positive, yellow = ignore.

Model and training. A U-Net with an EfficientNet-B4 encoder (ImageNet pretrained) takes 4-band RGB+NIR 512×512 chips. Sampling is 45% centred on water, 25% centred on hard negatives (roads, roofs, dark fields) and 30% random. The loss is weighted BCE + Dice, optimized with AdamW and a OneCycle schedule. Version 1 did not sample hard negatives and reached a test-area object F1 of only 0.59; version 2, with hard negatives, reached 0.83.

Inference and multiple years. Inference uses 1024-px sliding windows with horizontal-flip test-time augmentation. The output is thresholded at 0.5 and polygonised into objects with area, roundness and building/road proximity attributes. The 2018 and 2020 images were taken in autumn and are about 25% darker than 2022; inferring on them directly misses much dark clear water, so each tile's band means and variances are matched to the 2022 statistics first. The union of detections from the three years forms a "pond footprint" inventory, and the water area inside each footprint is computed per year. A pond counts as holding water in a year if its water area is at least 10% of its three-year maximum and at least 10 m^2.

3.2 Round tanks: two-stage detection

Why a different image source. A blob detector with CNN scoring on NAIP gave only 12–33% county-wide precision, because a tank is 3–13 pixels across at 0.6 m and indistinguishable from shrubs. On Google z19 imagery (~0.25 m) tanks are 10–50 pixels across and clearly visible.

Stage 1: YOLO11s candidate detection. 94 tank boxes were hand-labeled near TWDB livestock wells and bright NAIP blobs. Two active-learning rounds followed: the model was run on the training area, and high- and

medium-score detections were reviewed manually. This added 109 positives and 118 hard negatives (cattle, shrubs, bales, building shadows). The final dataset has 991 training and 140 validation images (384-px chips upscaled to 640) with 203 tank boxes. County-wide inference uses 384-px windows with 64-px overlap and cross-window NMS.

Stage 2: z20 type classifier. A z20 re-check showed that 46% of the single-stage high-confidence detections were round bales and 21% hay-ring feeders. A ResNet18 four-class classifier (water facility / hay-ring feeder / round bale / other) was therefore trained on 16 m chips at ~0.13 m. It used 443 manual labels (197 water, 108 bales, 29 feeders, 109 other). Five-fold cross-validation gave 84% overall accuracy, with water-class precision 0.90 and recall 0.85. Final output: YOLO confidence $\geqslant$ 0.2 and water probability $\geqslant$ 0.8.



Figure 4. The four candidate classes in stage 2 (z20, ~16 m view). Top to bottom: storage tanks and troughs (dark water or a metal shell inside, often with a shadow); hay-ring feeders (dark ring, light hay inside, trampled patches); round bales (uniform light disc, no trampling); other (shrubs, cattle, vehicles, sheds).

4. Models trained

#	Model • task	Architecture and input	Training data and settings	Validation	Status
1	**U-Net v2 (4-band) pond water segmentation**	**U-Net + EfficientNet-B4 (ImageNet pretrained); NAIP RGB+NIR, 512×512**	**v2 weak labels on 53 training tiles; continued from v1 for 20 epochs × 400 iterations, batch 16, BCE+Dice**	**val IoU 0.878**	**Final (ensembled with #9)**
2	U-Net v1 (4-band) pond water segmentation	Same	v1 weak labels, no hard-negative sampling; ~15 epochs	val IoU 0.90*	Superseded by v2
3	U-Net v2 (RGB only) pond water segmentation	Same, 3-band input	Same as #1	val IoU 0.816	Spare; not evaluated separately
4	ResNet18 (NAIP) tank candidate scoring	NAIP 4-band 48×48 chips	192 reviewed candidates near wells + 1,500 unlabeled candidates	CV AUC 0.89	Dropped (county precision 12–33%)
5	**YOLO11s v3 round-tank candidate detection**	**Google z19 RGB, 384-px chips upscaled to 640**	**203 tank boxes, 991 train / 140 val images; v1→v2→v3 successive training, early stopping**	**val mAP@0.5 ≈ 0.5**	**Superseded by #10**
6	**ResNet18 tank type classification**	**Google z20 RGB, 128 px → 224; 4 classes**	**443 manual labels, 30 epochs, class-weighted cross-entropy**	**CV accuracy 0.84 water P 0.90 / R 0.85**	**Superseded by #11**
Self-supervised upgrade (Section 7)					
7	Domain-SSL ViT-L (NAIP) self-supervised pretraining	DINOv3-sat ViT-L/16, input widened to RGB+NIR	Unlabeled county NAIP, 3 years; DINO + iBOT + KoLeo + Gram anchoring, 6,000 iterations	no labels	Backbone
8	Domain-SSL ViT-L (Google) self-supervised pretraining	DINOv3-sat ViT-L/16, RGB	520 k Google z19 county tiles, 7,000 iterations	no labels	Backbone
9	**ViT-L pond segmentation 2× pond water segmentation**	**DINOv3-sat ViT-L/16 + multi-level feature decoder + full-resolution detail branch; RGB+NIR, input upsampled 2×**	**Same weak labels as #1, 8,000 steps, batch 16**	**val IoU 0.900**	**Final (ensembled with #1)**
10	**ViT-L dense tank detector tank candidate detection**	**Same architecture, Google z19 RGB, outputs a tank-disc probability map**	**230 z20-verified water facilities + 333 hard negatives, 3,000 steps**	**val object F1 0.79**	**Final stage 1**
11	**ViT-L tank type classifier tank / feeder / bale / other**	**DINOv3-sat ViT-L/16 fine-tuned, z20 128 px → 224**	**584 z20 labels (including the dense detector's hard false positives)**	**CV water F1 0.825**	**Final stage 2**

*The v1 and v2 validation sets differ: the v2 set contains hard negatives (roads, fields, roofs), so its numbers are lower, but v2 is clearly better on the test area (Section 5). The YOLO validation set has only ~30 instances, so mAP fluctuates; the independent test in Section 5 is what counts.

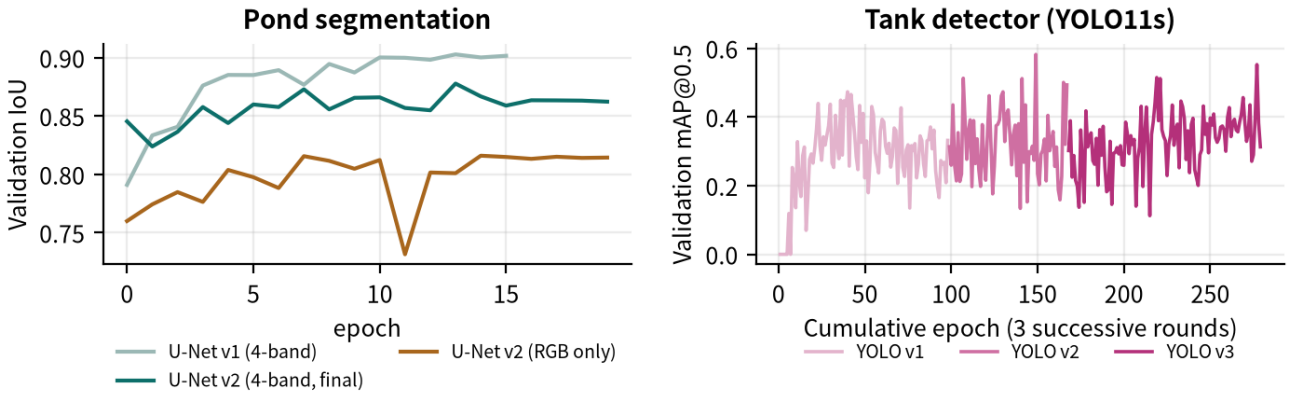


Figure 5. Training curves. Left: validation IoU of the pond segmentation models. Right: validation mAP@0.5 of the tank detector over the three successive active-learning rounds.

Weights: `runs/unet_rgbn_v2/best.pt`, `runs/unet_rgb_v2/best.pt`, `runs/yolo/tank_v3/weights/best.pt`, `runs/tank_type_cls.pt`.

5. Accuracy assessment

All evaluation is on the test area, which was never used for training or tuning. Because the weak labels are incomplete, the final numbers come from manual checks of individual images. Each sample was viewed in NAIP true colour, NIR false colour and Esri (or Google z20) high-resolution imagery. "Strict" counts unclear samples as errors; "lenient" counts them as correct.

5.1 Ponds

Evaluation (test area, NAIP 2022)	Precision	Recall	F1
vs weak labels: spectral-rule baseline	0.12	—	0.21
vs weak labels: U-Net v1	0.49	0.75	0.59
vs weak labels: U-Net v2 (pixel IoU 0.83)	0.92	0.76	0.83
Manual check: 60 random detections / 42 water-holding NWI ponds	**0.88–0.93**	**0.81**	**—**

Of 60 detections, 53 were real water, 4 were false positives (3 roads/railway, 1 dark roof) and 3 were uncertain. Of 60 random NWI ponds, 18 (30%) were dry in July 2022. The remaining 42 held water and the model found 34; the misses were small water under tree shade and ponds covered with algae or vegetation.

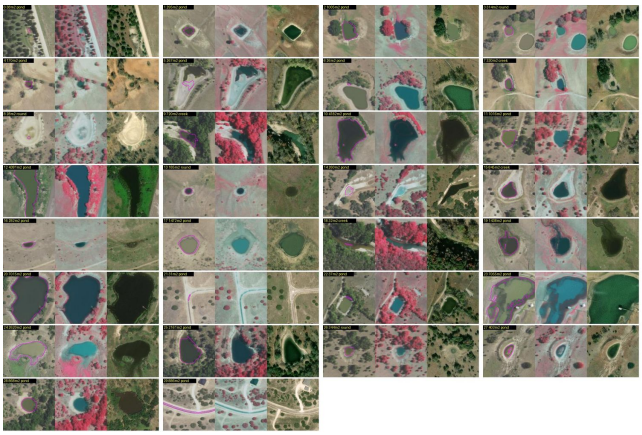


Figure 6a. Precision-check samples (part). Each group: NAIP true colour with the detection outline in magenta | NAIP NIR | Esri 2021.

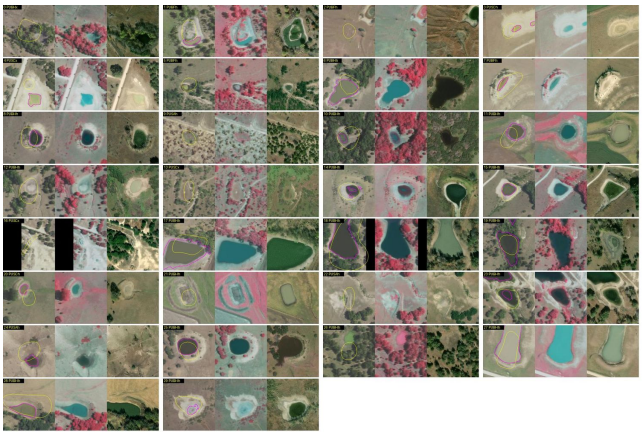


Figure 6b. Recall-check samples (part). Yellow: NWI pond; magenta: model detection. Water without a magenta outline is a miss.

5.2 Round tanks

System (test area)	Recall	Precision (strict–lenient)	Detections
NAIP 0.6 m blobs + CNN	—	12–33%	—
Google z19 YOLO, single stage (confidence ⩾ 0.35)	60%	29%	1,025
YOLO + z20 type classifier (confidence ⩾ 0.2, water probability ⩾ 0.8)	**72%**	**67–85%**	**741**

The recall reference is 18 water facilities picked independently in the test area and confirmed at z20; the system found 13 of them. Precision comes from a random sample of 33 final detections checked at z20. The remaining false positives are mostly vehicles, trailers, sheds and shrubs; misses are mostly tanks under tree cover or inside livestock pens.

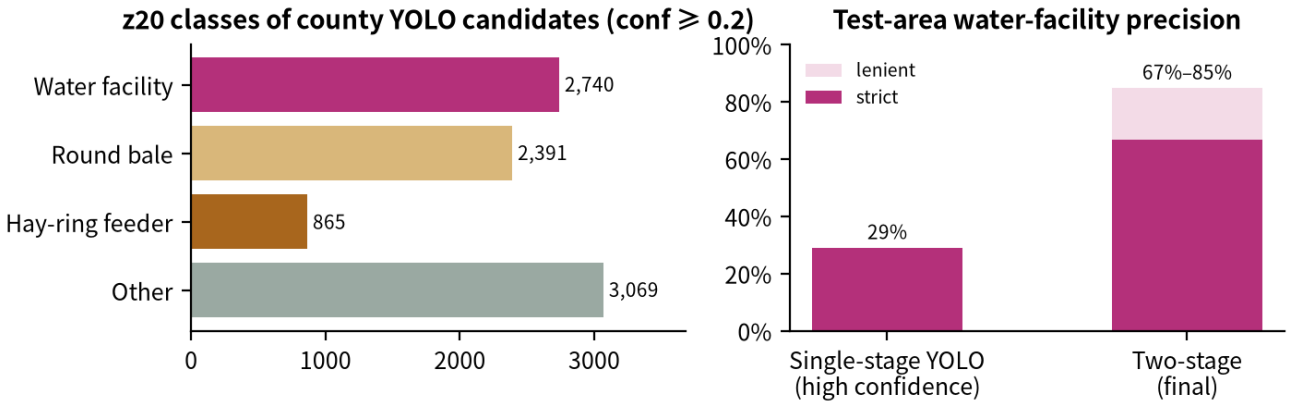


Figure 7. Effect of stage 2. Left: z20 classes of all county YOLO candidates; water facilities are less than a third (the final output also requires water probability ⩾ 0.8, which is why it has 1,849 rather than 2,740). Right: test-area water-facility precision, single stage vs two stages.

6. Results (original models)

6.1 County ponds and three years of water status

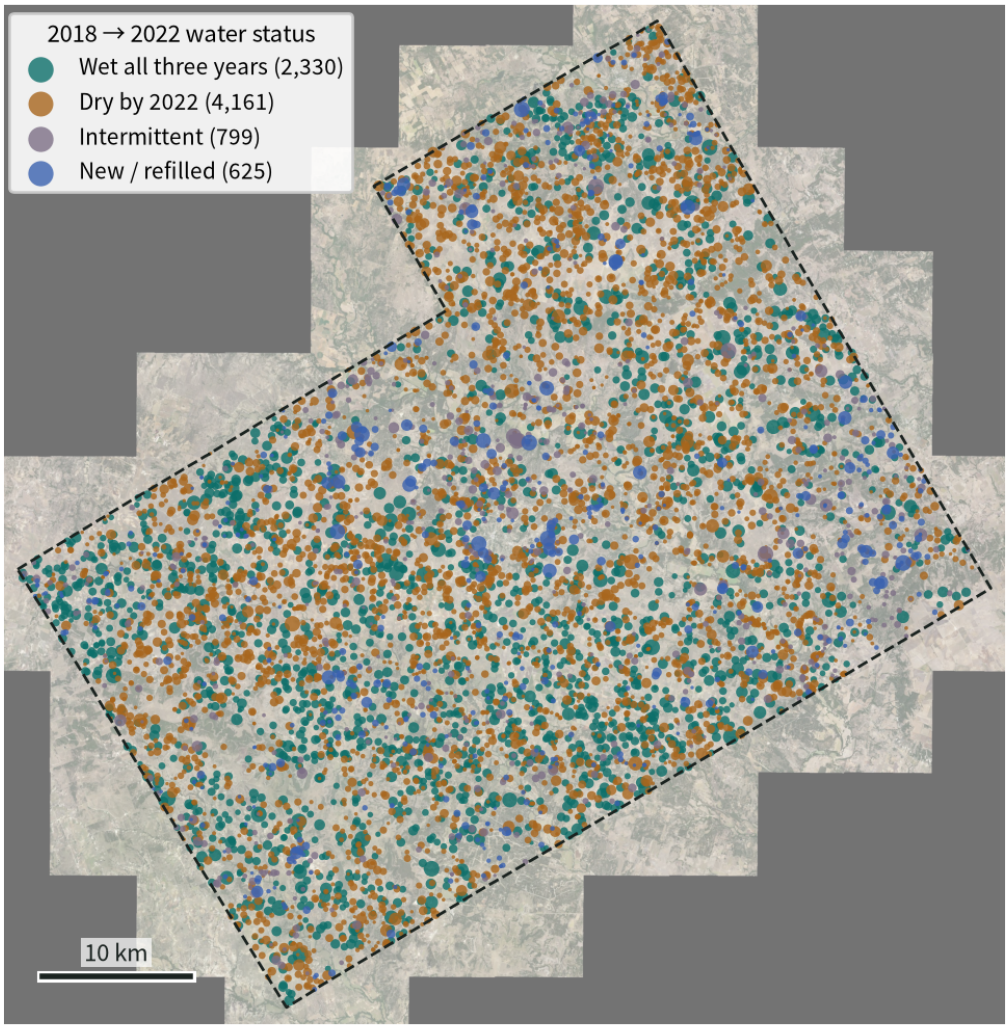


Figure 8. County pond footprints and 2018→2022 water status. Marker size scales with pond area. The 4,161 "dry by 2022" ponds are mostly small earthen ponds; since 2022 recall is about 81%, a few of them are missed water-holding ponds.

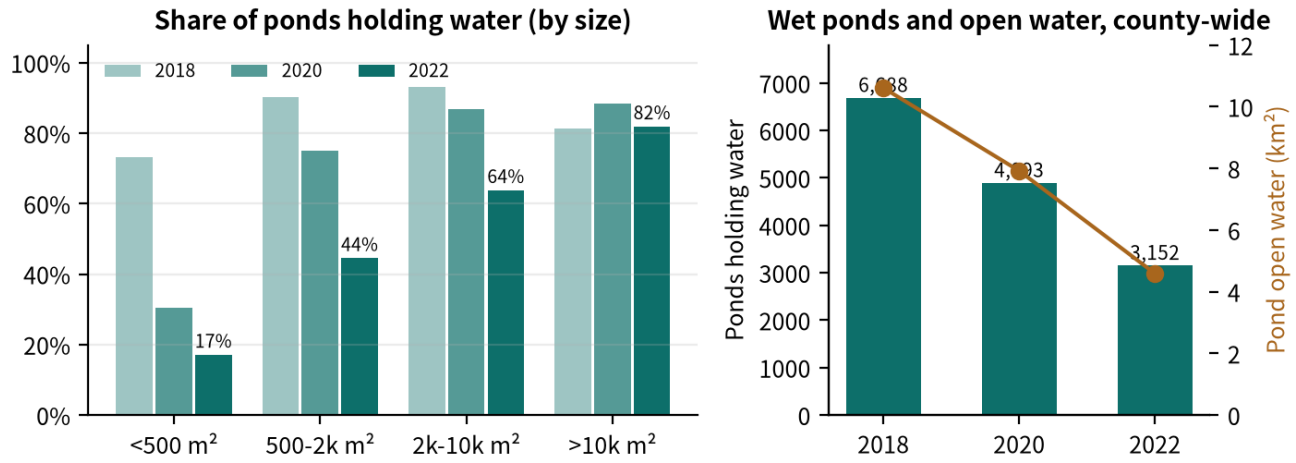


Figure 9. Per-year statistics. Left: share of ponds holding water in each size class; small ponds dry out first in drought years. Right: county ponds holding water (bars) and total pond open water (line), 10.6 → 7.9 → 4.6 km².



Figure 10. Cross-year detection examples. 24 random objects detected in 2020 but not in 2022; each group shows 2020 true colour | 2020 NIR | 2022 true colour | 2022 NIR. About 20 are ponds that held water in 2020 and were empty in 2022; the rest are roads or small shadows.

6.2 County round tanks and storage tanks

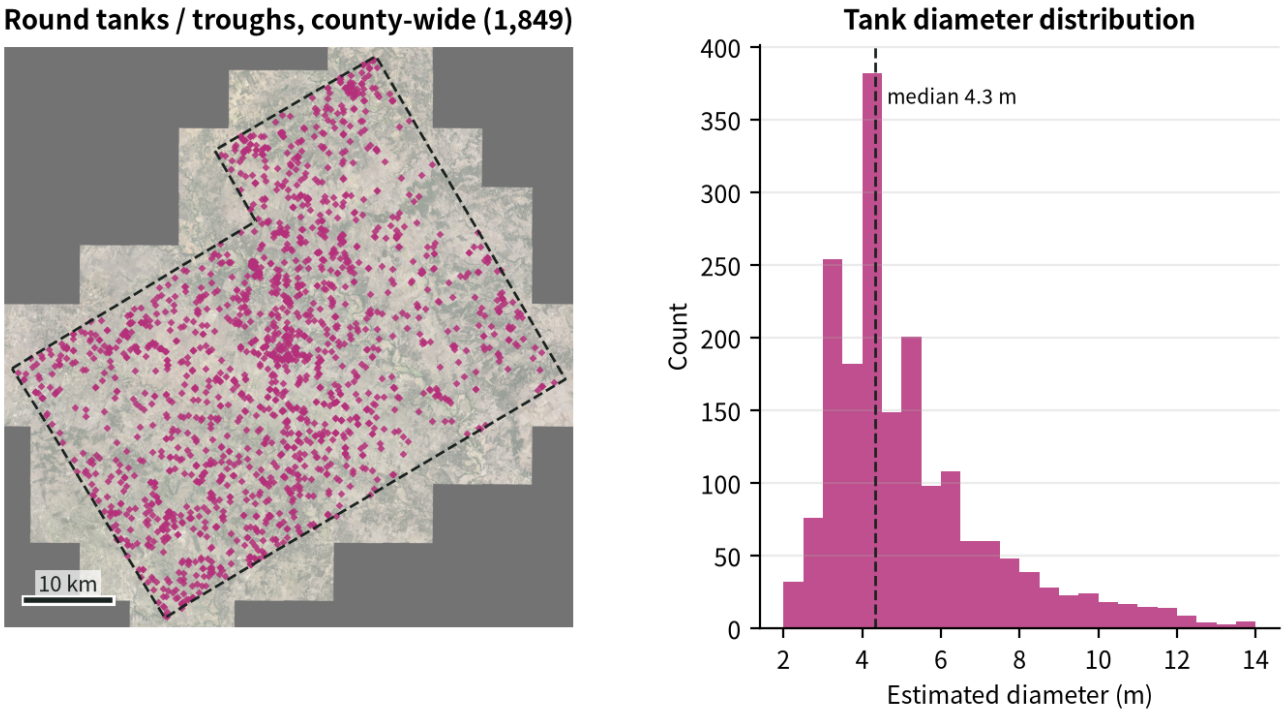


Figure 11. Distribution and diameter of 1,849 round tanks county-wide. About 0.84 per km^2. Median estimated diameter 4.3 m, interquartile range 3.8–6.1 m (box size includes ~0.4 m margin); those over 8 m are mostly concrete reservoirs or large metal storage tanks.



Figure 12. Manually confirmed water facilities in the test area (z20). Round troughs where cattle trails converge, concrete troughs holding water, metal and plastic storage tanks.



Figure 13. Ranch-scale example (1.2 km × 1.2 km). Cyan: pond water detected in NAIP 2022; magenta: tanks identified by the two-stage model.

7. Self-supervised upgrade

The models above all use convolutional networks pretrained with **supervision** on ImageNet (EfficientNet, ResNet18, YOLO11s). This chapter switches to the current approach: first learn general remote-sensing features with **label-free** self-supervised learning, then fine-tune with a small number of labels. Three things were done:

- **A public self-supervised satellite foundation model**: DINOv3-sat ViT-L/16 (300 M parameters, released 2025), trained with the DINOv3 method on 493 M Maxar satellite images without any human labels.
- **Domain-adaptive self-supervised pretraining on county data** (trained here): starting from DINOv3-sat, training continued on fully unlabeled county imagery. The NAIP branch widens the input to 4 bands so the model **learns to use near-infrared by itself** (DINOv3-sat has only seen RGB); the Google branch adapts to ranch-facility detail at 0.25 m.
- **Downstream models on the self-supervised backbone**: pond segmentation becomes ViT-L + multi-level feature decoder + full-resolution detail branch; tank detection becomes a ViT dense detector; tank type classification becomes a ViT classifier.

7.1 Self-supervised pretraining

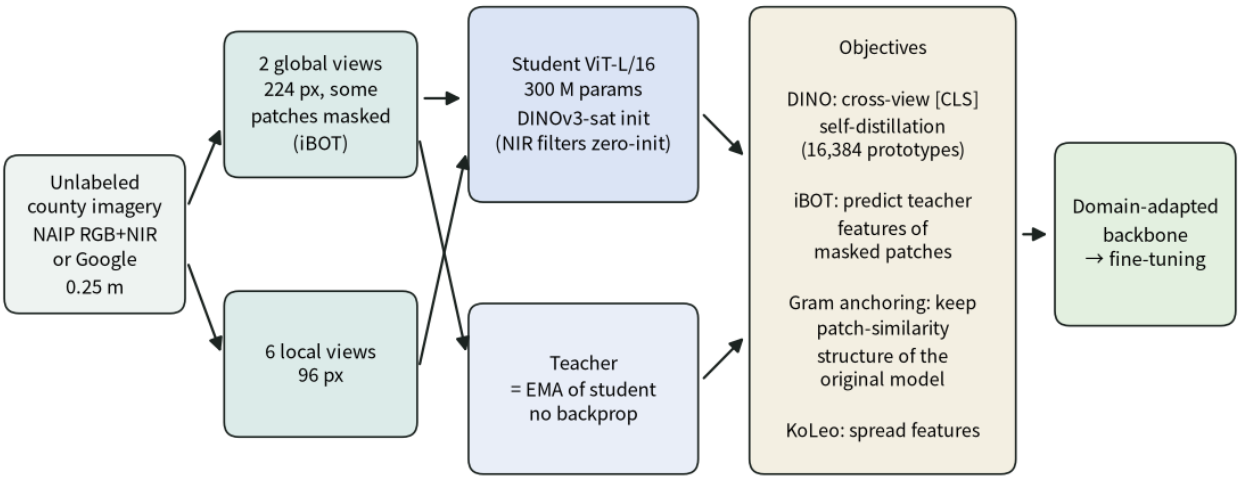


Figure 14. Domain-adaptive self-supervised pretraining. Each unlabeled image yields 2 global and 6 local views. The student learns to make its representations of all views agree with the teacher (an exponential moving average of the student) and to predict the teacher's features of masked patches. Gram anchoring keeps the patch-to-patch similarity structure close to the original DINOv3-sat, preventing feature degradation during continued training.

Setting	NAIP branch	Google branch
Unlabeled data	NAIP 2018/2020/2022, 231 county tiles (~100 GB), random 192 m windows	520 k county z19 tiles (~0.25 m), random 96 m windows
Input	RGB + NIR, 4 bands (NIR filters zero-initialised)	RGB
Objective	DINO ([CLS] self-distillation, 16,384 prototypes) + iBOT (masked-patch prediction) + KoLeo + Gram anchoring; EMA teacher (0.994→1)	
Amount of training	6,000 iterations × 96 images × 8 views ≈ 580 k images*	7,000 iterations × 96 images × 8 views ≈ 670 k images
Hardware / time	2 × A100 80 GB, ~3.5 h	2 × A100 80 GB, ~4 h
Outcome	No collapse (teacher entropy 3.4–4.0, stable feature variance); median relative weight change 1.6%; NIR filters grew from 0 to 7% of the RGB filter norm	No collapse; median relative weight change 1.4%

*7,000 iterations were planned. At ~6,500 a data-loading worker was killed by the out-of-memory killer; the checkpoint at iteration 6,000 is used, by which point the cosine learning-rate schedule had nearly reached its minimum.

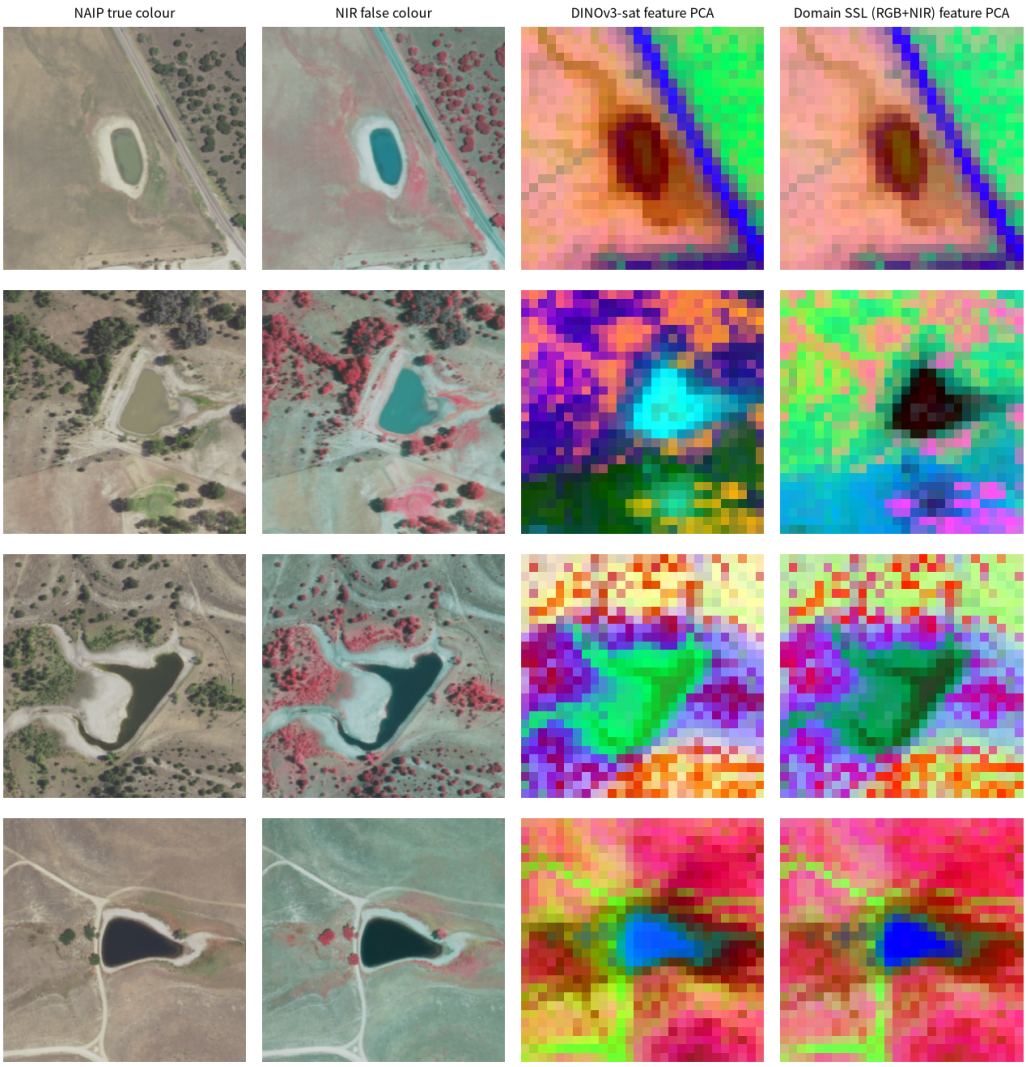


Figure 15. Self-supervised features (test area, no labels). The two right columns show the first three principal components of the patch features as false colour. Without seeing any labels, the model already separates pond water, exposed pond banks, woodland and roads into distinct feature clusters.

7.2 Pond segmentation results

All models use the same weak labels, sampling and number of training steps, and are evaluated on the same test area. "2×" means the image is upsampled 2× before entering the ViT: at 0.6 m a 16-pixel patch covers 9.6 m × 9.6 m, larger than many small ponds; after upsampling each token covers 4.8 m.

Model (test area, NAIP 2022)	Pretraining	Pixel IoU	Obj. precision	Obj. recall	F1	NWI recall
U-Net v2 EfficientNet-B4 (original)	ImageNet supervised	0.829	0.915	0.764	0.833	0.81
ViT-L, RGB only, 1×	DINOv3-sat SSL	0.805	0.901	0.648	0.754	0.81
ViT-L, RGB+NIR, 1×	DINOv3-sat SSL	0.884	0.978	0.658	0.787	0.74
ViT-L, RGB+NIR, 1×	**+ county domain SSL**	0.883	0.967	0.697	0.810	0.74
ViT-L, RGB+NIR, 2×	DINOv3-sat SSL	0.898	0.932	0.764	0.840	0.76
ViT-L, RGB+NIR, 2×	**+ county domain SSL**	0.881	0.901	0.767	0.829	0.79
Ensemble: U-Net v2 + ViT-L 2× (final)	**mean of probabilities**	**0.905**	**0.943**	**0.766**	**0.845**	**0.81**

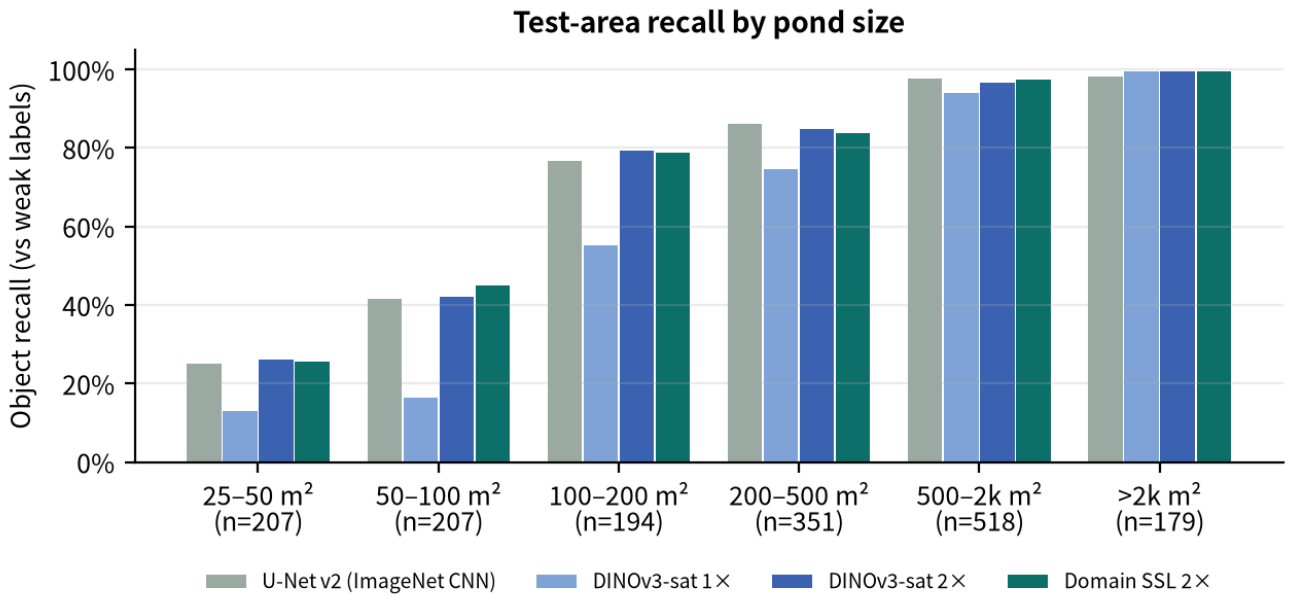


Figure 16. Recall by pond size. At 1× the ViT misses many ponds under 500 m² (one patch is already 92 m²). With 2× input, small-pond recall recovers to the CNN's level while keeping the ViT's more accurate boundaries (pixel IoU 0.90) and fewer false positives.

A manual check of 60 random detections from the 2× ViT found 51 real ponds, 5 errors and 4 uncertain, i.e. 85% (strict) to 92% (lenient) precision, comparable to U-Net's 88–93%. Three of the 5 errors are backyard swimming pools, which the existing "near building" flag can filter out.

Conclusions:

- The gains of the self-supervised ViT come mainly from three things: the DINOv3-sat features, the near-infrared band, and the 2× input.

- With all training data, county domain SSL adds +0.023 F1 in the 1× setting (mainly +0.04 small-object recall) and nothing further at 2×.
- The CNN + ViT ensemble is best on every metric and is used as the final pond model.
- **County domain SSL matters most when labels are scarce**: trained on only 10% of the imagery (5 tiles), the domain-SSL ViT 2× reaches F1 0.831 and IoU 0.897. This essentially matches the U-Net trained on all data (0.833 / 0.829) and beats U-Net and DINOv3-sat trained on the same 10% (table below).

Trained on 10% of the imagery (5 tiles)	Pixel IoU	Obj. precision	Obj. recall	F1	NWI recall
U-Net v2 (ImageNet supervised)	0.827	0.910	0.740	0.816	0.81
ViT-L 2× (DINOv3-sat SSL)	0.883	0.812	0.825	0.819	0.86
ViT-L 2× (+ county domain SSL)	**0.897**	**0.892**	**0.778**	**0.831**	**0.81**
Reference: U-Net v2 on all 53 tiles	0.829	0.915	0.764	0.833	0.81

Pond weak labels are generated automatically and are plentiful, so with all data the three models differ little. The 10% experiment simulates having only a few labeled images (for example, moving to a region without NWI).

7.3 Round-tank results

Label efficiency of the type classifier. Same 443 manual labels, same 5 folds; only the backbone and the number of labels change:

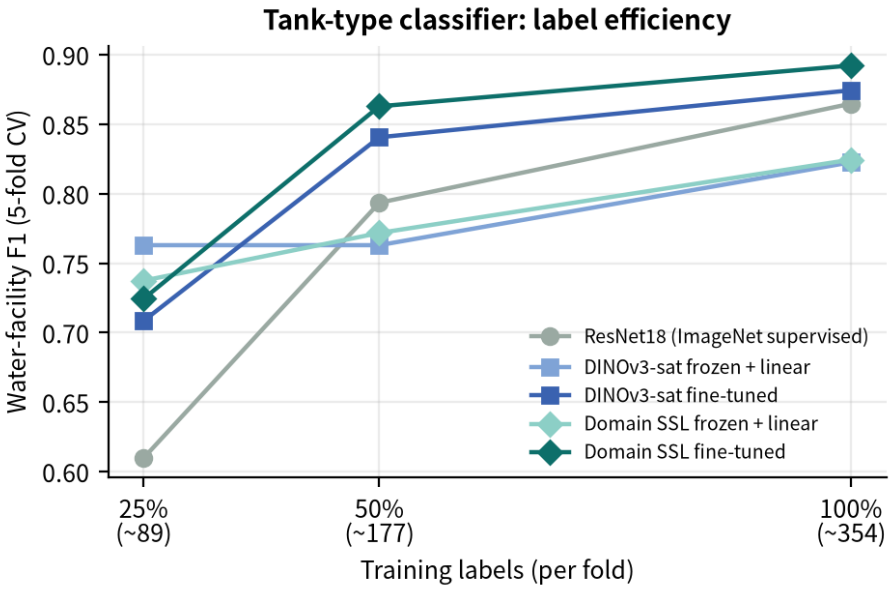


Figure 17. Water-facility F1 of the tank type classifier. With ~90 labels, self-supervised features (frozen + linear) beat the ImageNet ResNet18 by 0.13–0.15. With all labels, the fine-tuned county domain-SSL model (Google branch) is best (0.892 vs ResNet18 0.865).

A dense detector replaces YOLO. The original two-stage system's recall was capped by YOLO at 72%. The new detector outputs a tank-disc probability map directly from the ViT-L backbone and is followed by the ViT type classifier. Its training data are: 230 z20-verified water facilities as clean positives; 333 look-alike hard negatives (bales, feeders, sheds, shrubs); and a second active-learning round, in which the dense detector was run on the training area and its detections were labeled manually.

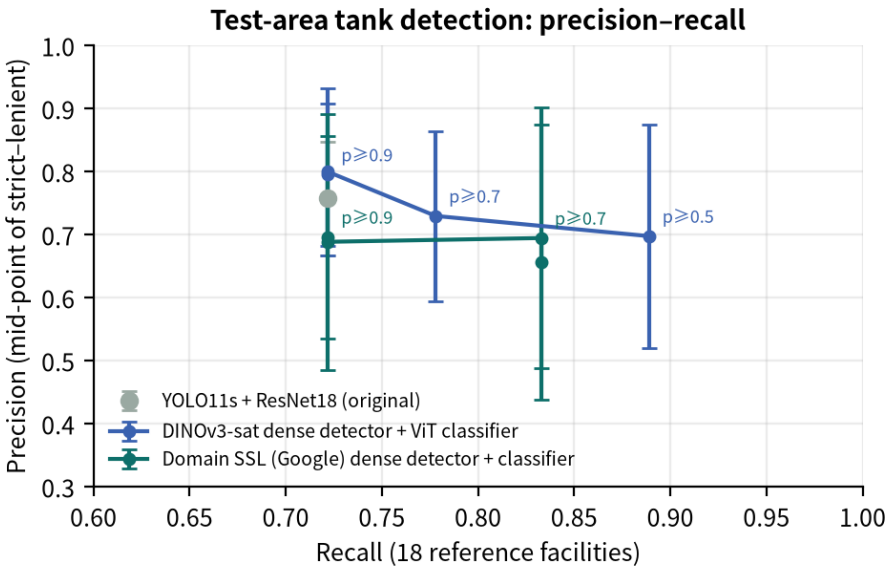


Figure 18. Test-area tank detection, precision vs recall. Error bars span strict–lenient precision. The self-supervised system can keep similar or higher precision at the same recall (72%) while finding more facilities, or raise recall to 89%.

Tank system (test area)	Recall (18 refs)	Precision (strict–lenient)	Detections	Est. real facilities
YOLO11s + ResNet18 (original)	72%	67–85%	741	~490–630
DINOv3-sat dense detector + ViT classifier, p ⩾ 0.8	72%	67–93%	939	~630–880
DINOv3-sat dense detector + ViT classifier, p ⩾ 0.7 (final)	**78%**	**60–87%**	**1,096**	**~650–950**
DINOv3-sat dense detector + ViT classifier, p ⩾ 0.5	89%	52–88%	1,337	~700–1,170
County domain SSL (Google) dense detector + domain-SSL classifier, p ⩾ 0.7	83%	49–90%	1,034	~500–930

Precision comes from 48 randomly sampled detections per system, checked manually at z20; higher thresholds leave fewer samples (22–48), so precision is uncertain by about ±12–15 percentage points. The reference set has only 18 facilities, so each one is 5.6 percentage points of recall.

Conclusion: with the self-supervised ViT dense detector, about 30% more real facilities are found at equal recall, or recall can be raised from 72% to 78–89%. Domain SSL on Google tiles gives a small gain for tank **classification** and no measurable gain for **detection**.

7.4 Final models

Task	Final model	vs original
Pond water segmentation	Ensemble: U-Net v2 (EfficientNet-B4) + DINOv3-sat ViT-L/16 2× (RGB+NIR, self-supervised pretraining)	F1 0.833 → 0.845; IoU 0.829 → 0.905; precision 0.915 → 0.943; recall unchanged
Tank detection	DINOv3-sat ViT-L/16 dense detector (second active-learning round)	Recall 72% → 78–89%
Tank type classification	DINOv3-sat ViT-L/16 fine-tuned (584 z20 labels)	~+30% real facilities at equal recall
Self-supervised backbones (reusable)	County domain-SSL ViT-L: NAIP 4-band version, Google 0.25 m version	Tank classification F1 (all labels) 0.865 → 0.892; ponds 1× F1 +0.023; ponds with 10% data F1 0.831, matching full-data U-Net

7.5 County results regenerated with the new models

The maps and statistics in Section 6 come from the original models. Below are the county results rerun with the final self-supervised models (ponds: ensemble, three NAIP years; tanks: ViT dense detector + ViT classifier, $p \geqslant 0.7$). To save compute, the ViT was run only on 512-px windows where the U-Net probability exceeds 0.05 or NIR-water pixels exist; in other windows the ensemble could not detect water anyway.

County (inside boundary)	Original models	Self-supervised models
Pond footprints	7,915	7,707
Ponds holding water 2018 → 2020 → 2022	6,688 → 4,893 → 3,152	6,735 → 4,829 → 3,072
Pond open water 2018 → 2020 → 2022 (km^2)	10.6 → 7.9 → 4.6	12.0 → 8.3 → 4.6
"Intermittent" / "new or refilled"	799 / 625	647 / 523
Round tanks / storage tanks	1,849	3,026

Pond counts and the wet/dry trend agree with the original model. The new model detects about 1.4 km^2 more water in 2018 (more complete boundaries), and about 20% fewer ponds flip between classes across years ("intermittent", "new or refilled"), indicating fewer false changes caused by false positives and misses. The tank count rises by about 64%, consistent with the test-area result of ~48% more detections and ~30% more real facilities at equal recall.

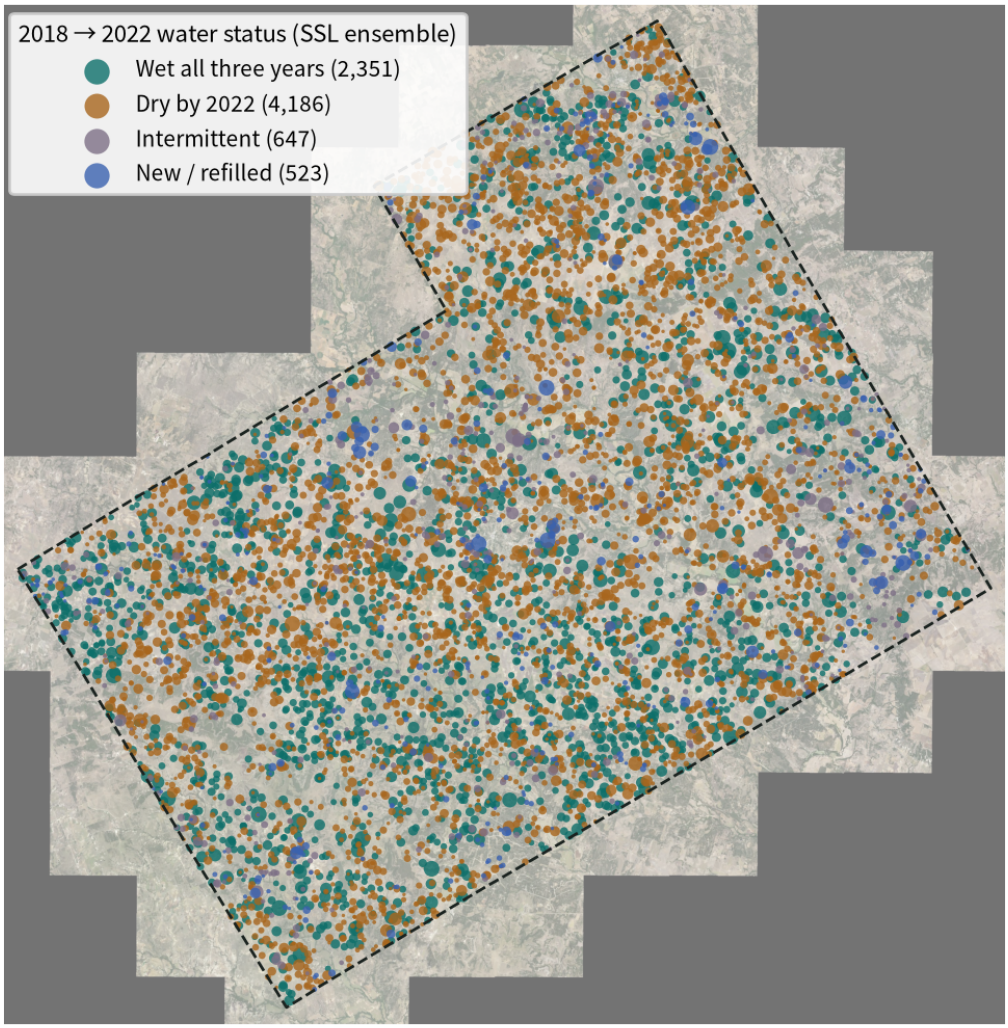


Figure 19. County pond footprints and 2018→2022 water status (self-supervised ensemble).

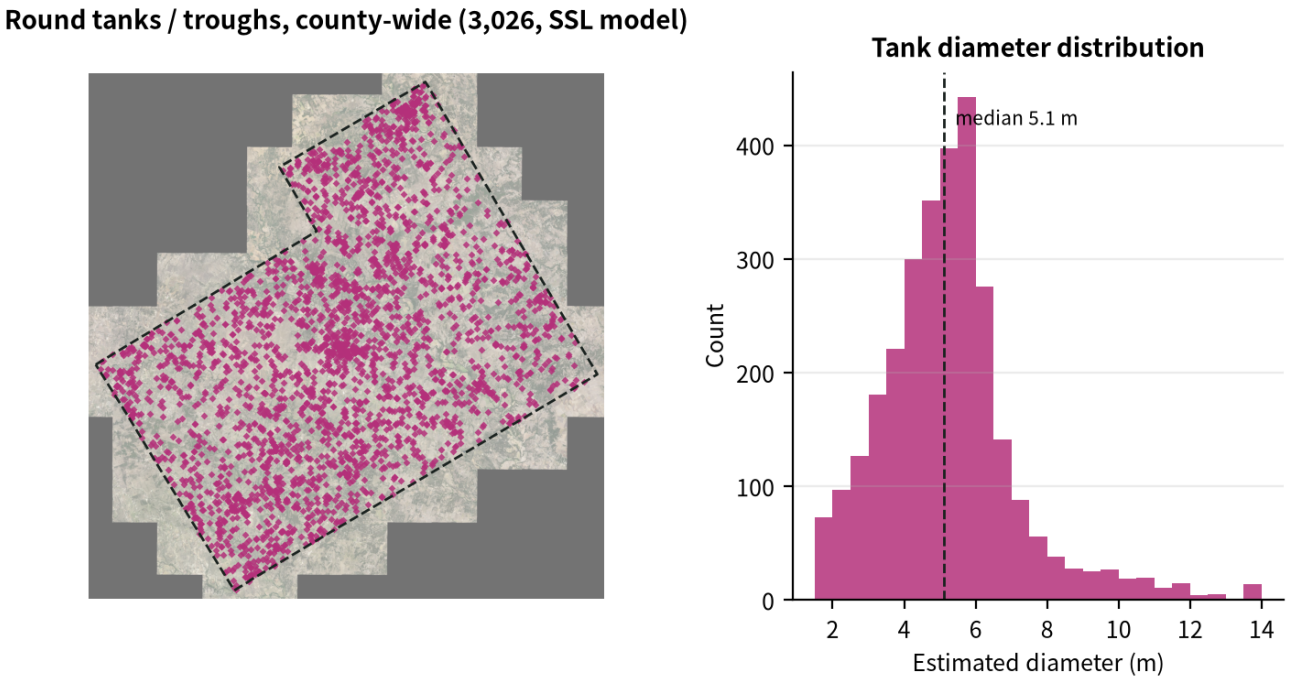


Figure 20. 3,026 round tanks / storage tanks county-wide (self-supervised model) and their diameters. Diameter is derived from the detected disc area; median 5.1 m.

8. Limitations and next steps

- **Evaluation sample size.** 60 checked ponds per test, 18 tank recall references and 22–48 precision samples give uncertainties of about ±10–15 percentage points. Firm conclusions need larger verification sets, ideally with field confirmation from ranchers.
- **Google imagery.** Acquisition dates are inconsistent and unpublished, so it is suitable for locating tanks but not for judging whether they held water on a given day; its terms of use restrict bulk download. A production project should use licensed or public high-resolution orthoimagery (e.g. TxGIO StratMap, after confirming county coverage and year) or drones.
- **Failure modes.** Ponds: algae or vegetation cover and small water under trees; about 200 local pond labels should help. Tanks: tree cover and tanks inside pens; 100–200 more z20 samples (including vehicle, trailer and shed negatives) should help.
- **Revisit frequency.** NAIP is flown every two years, which is not enough for seasonal tracking. Large ponds can be moved to Sentinel-2 (10 m, 5-day) or PlanetScope (3 m, daily); whether a tank holds water needs regular ⩽ 0.3 m imagery or ground water-level sensors.
- **Per-ranch output.** With ranch boundaries from the ranchers, per-ranch inventories of ponds and tanks with year-by-year water status can be produced directly.

Appendix: output files

File (relative to `/home/ychen289/whe/pond`)	Contents
`outputs/vector/water_sources_ssl_{2018,2020,2022}.gpkg`, `outputs/pond_inventory_multiyear_ssl_hamilton.csv`	Self-supervised ensemble: water detections for three years and per-pond water status (recommended)
`outputs/vector/round_tanks_hamilton_ssl.geojson`	3,026 round tanks from the self-supervised model (recommended), with detection confidence and four class probabilities
`outputs/vector/water_sources_2022_hamilton.geojson`, `outputs/vector/water_sources_{2018,2020,2022}.gpkg`	Original model: water detections with area, roundness, class and false-positive flags
`outputs/pond_inventory_multiyear_hamilton.csv`	Original model: 7,915 pond footprints with water area per year
`outputs/vector/round_tanks_hamilton_final.geojson`	Original model: 1,849 round tanks
`outputs/prob/{year}/`, `outputs/prob_vit/{year}/`, `outputs/prob_ens/{year}/`	Per-pixel water probability maps (U-Net, ViT, ensemble)
`runs/ssl/naip/ckpt.pt`, `runs/ssl/google/ckpt.pt`	County domain-SSL ViT-L backbones (NAIP 4-band / Google RGB)
`runs/seg_dinov3sat_rgbn_x2/`, `runs/tankdet_dinov3sat_v2/`, `runs/tank_type_vit_dinov3.pt`	Final ViT pond segmentation, dense tank detector and tank type classifier weights
`outputs/eval/`, `outputs/eval/ssl/`, `data/manual/`	All evaluation results, manual labels and verification records
`README.md`, `scripts/`	Full reproduction workflow (scripts 01–29 and ssl_*.py)